

BUSINESS & INNOVATION

1. Market Analysis

Terralink is a **portable LoRa-based ad-hoc emergency communication system** designed to provide communication in locations where conventional communication infrastructure is unavailable, unreliable or disrupted. The system uses portable Terralink nodes to form a temporary communication network, allowing emergency information to be exchanged between users without depending entirely on fixed communication towers.

The need for reliable communication becomes particularly important in environments such as **hilly regions, forests, trekking routes, rural areas and disaster-affected locations**. In such areas, cellular connectivity may be weak or completely unavailable. During an emergency, the inability to communicate the person's condition and location can make rescue operations more difficult.

Terralink addresses this requirement by providing a portable communication system that can be carried and deployed according to the situation. When a person in distress activates the SOS function, emergency information and location data can be transmitted through available Terralink nodes. Rescuers carrying their own Terralink nodes can participate in the same network while moving through the search area.

Target Market

The potential market for Terralink includes organizations and individuals who operate in areas where conventional communication may not always be reliable. The major target users include:

- Search and rescue teams
- Disaster management organizations
- Trekking and adventure groups
- Forest and wildlife departments
- NGOs and humanitarian organizations
- Rural and remote communities
- Field survey and research teams
- Industrial and mining operations
- Outdoor event organizers

The product can serve both **individual users and institutional customers**. Individual users may require the system for trekking, adventure activities or remote

travel, while organizations can deploy multiple units for rescue, field operations or emergency response.

Market Need

Existing communication systems generally depend on cellular towers or other communication infrastructure. When this infrastructure is unavailable because of geographical limitations, network failure or disaster conditions, communication becomes difficult.

Terralink is positioned as a **local, portable and infrastructure-independent communication solution** for such situations. Its ad-hoc network approach allows multiple nodes to participate in communication, making it suitable for temporary deployment.

The product also has potential applications beyond emergency communication. The same network concept can be adapted for remote field operations, temporary events, industrial sites and other environments where portable communication is required.

2. Cost Estimation

Terralink has been developed as a **compact PCB-based product**, with the major electronic functions integrated onto fabricated printed circuit boards and enclosed within a portable housing.

The hardware development involves schematic design, PCB layout, component placement, routing and generation of Gerber manufacturing files. The Gerber files are provided for PCB fabrication, after which the fabricated boards are assembled with the required components to form the product.

The major functional sections of the product include the **microcontroller, LoRa communication, GPS, SOS interface and power-management circuits**.

The target selling price of the product is approximately:

₹1,500–₹2,000 per unit

The estimated cost distribution is shown below:

Component / Section	Estimated Cost
Microcontroller	₹200–₹300
LoRa communication section	₹200–₹350
GPS section	₹250–₹400

Component / Section	Estimated Cost
Battery & power management	₹250–₹350
PCB & connectors	₹150–₹250
SOS/interface components	₹50–₹100
Enclosure & assembly	₹150–₹250
Miscellaneous	₹100–₹150
Approximate Total	₹1,500–₹2,000

The actual manufacturing cost can vary depending on component sourcing, PCB fabrication charges, production quantity, assembly requirements and enclosure selection.

For larger production volumes, the unit cost can potentially be reduced through **bulk procurement, standardized components, optimized PCB manufacturing and efficient assembly processes**.

The integration of the electronic circuits onto fabricated PCBs also provides a more organized and scalable hardware solution compared with systems assembled using separate development boards and extensive external wiring.

3. Business Model

Terralink follows a combination of **direct sales and rental services**. This model allows the product to serve customers who require permanent access as well as users who need the system only for a limited period.

3.1 Direct Sales

Terralink units can be sold directly to individuals and organizations that require regular access to emergency communication.

Customers can purchase multiple units depending on their team size and operational requirements. For example, a trekking organization can deploy a set of Terralink units among its guides and participants, while a rescue organization can provide units to its field personnel.

Institutional customers can also receive customized packages based on the number of units required and their specific operational requirements.

3.2 Rental Services

The rental model is intended for customers who require Terralink for a temporary activity.

For example, a trekking group conducting a short expedition may require several communication units only for the duration of the expedition. Instead of purchasing the complete system, the organization can rent the required number of Terralink units and return them after the activity.

Potential rental customers include:

- Trekking and adventure groups
- Outdoor event organizers
- Disaster-relief teams
- Search and rescue teams
- Field survey teams
- Research expeditions
- Temporary industrial operations

Rental packages can be offered on a **daily, weekly or event-based basis**. This reduces the initial investment for temporary users and provides an additional recurring revenue opportunity.

3.3 Additional Revenue Opportunities

Additional revenue can be generated through services such as:

- Maintenance and repair
- Battery replacement
- Firmware updates
- Technical support
- Replacement of damaged units
- Customized institutional deployments

This provides multiple possible revenue streams while keeping the core product simple.

4. Commercialization Strategy

Terralink can be commercialized through a combination of **direct sales, rental deployment and institutional partnerships**.

Direct Marketing and Sales

The product can initially be introduced to trekking organizations, adventure groups, NGOs, emergency-response teams, research organizations and industries operating in remote areas.

Demonstrations and controlled field trials can be conducted to show the practical operation of Terralink. These demonstrations can help potential customers understand the advantages of portable ad-hoc communication.

Rental Deployment

A dedicated rental fleet can be maintained for customers who require Terralink for short-term activities. This approach can be particularly useful for trekking expeditions, outdoor events, temporary field surveys and emergency operations.

The rental model can also help introduce the product to new customers, as organizations can evaluate the system during an actual activity before deciding whether to purchase it.

Institutional Partnerships

Partnerships can be established with organizations such as:

- Disaster management organizations
- Forest departments
- NGOs
- Trekking organizations
- Industrial companies
- Emergency-response teams
- Research institutions

Institutional partnerships can support larger-scale deployment and provide opportunities for field testing under different operating conditions.

Market Expansion

The initial market can focus on **trekking, adventure activities, field operations and emergency-response applications**. After establishing the product in these segments, the system can be expanded into industrial, rural communication, disaster-management and other specialized applications.

5. Market Pitch

Terralink – Communication Without Fixed Towers

Terralink is a portable **LoRa-based ad-hoc emergency communication system** that enables temporary communication between users in areas where conventional communication infrastructure may be unavailable or unreliable.

When a person in distress activates the SOS function, emergency information and location data can be transmitted through available Terralink nodes. Rescuers equipped with their own nodes can move through the affected area while participating in the communication network.

Key Product Benefits

- Portable and infrastructure-independent
- LoRa-based long-range communication
- Ad-hoc multi-node networking
- SOS emergency functionality
- GPS-based location sharing
- Battery-powered operation
- Compact fabricated-PCB implementation
- Suitable for both purchase and rental

The combination of these features makes Terralink suitable for emergency communication in remote and challenging environments.

Market Statement

“Terralink enables temporary communication between victims and rescuers without requiring fixed communication towers.”

The product focuses on providing a practical communication link where conventional cellular communication may not be dependable.

6. Future Scope

Although Terralink has been developed as a functional PCB-based product, future versions can introduce additional hardware and software capabilities to improve network performance, reliability and usability.

Dynamic Network Management

Future versions can implement improved automatic node discovery and network management. The system can identify available nodes and establish communication paths as users move through the operating area.

This can help maintain connectivity when nodes change position or when some nodes become unavailable.

Intelligent Routing

Advanced routing algorithms can be introduced to select suitable communication paths based on parameters such as **signal quality, node availability, distance and battery level**.

This can improve the reliability and efficiency of the network, particularly when several nodes are operating simultaneously.

Mobile Application

A companion mobile application can be developed to provide features such as emergency alerts, location visualization, network status and information about active nodes.

The application could provide rescuers and administrators with a convenient interface for monitoring emergency situations.

Extended-Power Operation

Future versions can incorporate improved power-management techniques and optional **solar-assisted charging** to support longer field deployments.

This can be useful for applications where access to conventional charging facilities is limited.

Additional Sensors

Environmental sensors such as temperature, humidity, smoke and gas sensors can be integrated in future versions.

This can expand Terralink from an emergency communication system into a broader platform for **remote monitoring and environmental sensing**.

Drone-Assisted Communication

A Terralink-compatible node mounted on a drone could be investigated for difficult terrain. By positioning a communication node at an elevated location, network coverage could potentially be extended in areas where ground-based nodes have limited connectivity.

7. Patent / Research Scope

The core technologies used in Terralink, including **LoRa, GPS and microcontrollers**, are established technologies. Therefore, patent protection would depend on the novelty and technical uniqueness of the specific implementation rather than the use of these technologies individually.

Potential areas for intellectual-property investigation include:

- Portable emergency ad-hoc network architecture
- Emergency-priority routing
- Dynamic routing for mobile nodes
- Rescuer-assisted network formation
- Energy-efficient emergency communication
- Integration of SOS and location information
- Temporary infrastructure-independent communication

A **prior-art and patentability search** should be conducted before pursuing patent protection to determine whether the specific implementation contains sufficient novelty and inventive contribution.

Research Scope

Terralink can also provide a platform for research in areas such as:

- LoRa-based ad-hoc networking
- Multi-hop communication
- Mobile-node routing
- Low-power wireless communication
- Emergency communication systems
- Network reliability
- Disaster-response communication
- Infrastructure-independent networking

Future research can focus on improving routing efficiency, network reliability, power consumption and communication performance under different terrain and deployment conditions.

Overall Business Opportunity

Terralink combines a **fabricated PCB-based communication product with a flexible purchase and rental business model**. The product addresses the need for temporary communication in areas where conventional communication infrastructure may not be available.

The proposed price range of **₹1,500–₹2,000 per unit** provides an affordable entry point, while rental services allow temporary users to access the system without purchasing multiple units.

The initial target market can include trekking groups, rescue teams, NGOs, outdoor organizations and field teams, followed by expansion into disaster management, industrial operations, rural applications and other specialized sectors.

With further improvements in dynamic networking, routing, power management, mobile applications and sensing capabilities, Terralink has the potential to develop into a broader platform for **emergency and remote-area communication**.